

Zhongweiyang Xu

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EDUCATION

- **University of Illinois Urbana-Champaign** Champaign, IL
MS/PhD Student in Electrical and Computer Engineering; GPA: 4.00 Aug. 2021 – May 2026
- **University of Illinois Urbana-Champaign** Champaign, IL
Bachelor of Science in Computer Engineering; GPA: 3.97 Aug. 2018 – May. 2021

RESEARCH OVERVIEW AND INTEREST

- **Generative Audio/Speech Modeling:** I'm interested in general audio/speech generation (SpeechLLMs, Diffusion/Flow Generative Models), and AudioLLMs for general audio/speech understanding. I'm also interested in solving multi-channel speech enhancement, separation, and dereverberation using diffusion generative models. I had experience with neural audio/speech codecs, and spatial audio generation/processing.

INDUSTRY EXPERIENCE

- **Tencent America** Bellevue, US
Research Scientist - Speech & Audio Understanding (Speech Generation) June 2026 - Current
 - **Audio Understanding:** Work on Hunyuan audio language model for paralinguistic speech understanding and general sound understanding.
- **Meta Reality Lab Research** Redmond, US
Part-time Student Researcher Aug 2025 - Dec 2025
 - **Preference-aligned Speech Enhancement:** Work on flow based speech enhancement with preference alignment for verifiable rewards including PESQ, STOI, WER, DNSMOS, UTMOS. We try approaches including DPO, Flow-GRPO, and DiffusionNFT to further optimize a pre-trained flow model.
 - **Generalized ArrayDPS:** Work on generalized ArrayDPS for simultaneous speech enhancement, separation, and dereverberation. We use a pre-trained speech diffusion model to refine any discriminative models, in an unsupervised, generative, and array-agnostic manner.
- **Meta Reality Lab Research** Redmond, US
Research Scientist Intern May 2025 - Aug 2025
 - **ArrayDPS++:** We propose to use a pre-trained diffusion model for unsupervised multi-channel speech enhancement. When generating the speech, we modify the diffusion trajectory by adding a likelihood guidance into the prior, which enables posterior sampling for enhancement. We propose to use a discriminative model to estimate spatial noise covariance for likelihood estimation.
- **Meta Reality Lab Research** Redmond, US
Research Scientist Intern Aug 2023 - Dec 2023
 - **FoVNet:** Propose a system for configurable Field-of-View speech enhancement on a smart glass with extremely low computation and distortion. The system uses beamformers as features, and tries to extract target speech. Then we use an adaptive multi-channel wiener beamformer for low-distortion processing.
- **Tencent AI Lab** Bellevue, US
Research Scientist Intern May 2023 - Aug 2023
 - **SpatialCodec:** Research on multi-channel speech coding. Propose SpatialCodec to code spatial and spectral information independently with Residual VQ-VAE.
 - **uSee:** Research on text-based speech enhancement with score diffusion.
 - **CodecSE:** Research on codec based generative speech enhancement.

- **Tencent Ethereal Lab** Shenzhen, China
Research Scientist Intern May 2022 - Aug 2022
 - **KiNN**: Develop a Knowledge-inspired **computationally efficient** model for **real-time speech enhancement**. This model is designed for deployment in the **Tencent Meeting**'s (most popular video conferencing app in China) speech AI codec.

RESEARCH AND TEACHING EXPERIENCE

- **UIUC SiNRG Lab** Champaign, IL
Graduate Student Advised by Prof. Romit Roy Choudhury May 2021 - May 2026
 - **Audio/Speech Inverse Problems**:
 1. Combine signal processing and modern machine learning to solve audio inverse problems like source separation, system identification, speech enhancement, DoA estimation, Beamforming...
 - **Generative AI**:
 1. Apply generative models like VQ-VAE, GAN, Flow, Diffusion, EBM, Autoregressive Language Model on signal of interest.
- **UIUC Coordinate Science Laboratory** Champaign, IL
Graduate Research Assistant August 2024 - Dec 2024
- **UIUC Coordinate Science Laboratory** Champaign, IL
Graduate Research Assistant May 2024 - August 2024
- **UIUC Coordinate Science Laboratory** Champaign, IL
Graduate Research Assistant August 2022 - May 2023
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 374 Teaching Assistant August 2020 - Dec 2020
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 101 Teaching Assistant Jan 2026 - May 2026
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 498/598 Teaching Assistant August 2025 - Dec 2025
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 434 Teaching Assistant Jan 2025 - May 2025
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 434 Teaching Assistant Jan 2024 - May 2024
- **UIUC Electrical and Computer Engineering** Champaign, IL
ECE 374 Teaching Assistant Jan 2021 - May 2021

PUBLICATIONS

- **ICASSP 2026**: Zhongweiyang Xu, Ashutosh Pandey, Juan Azcarreta, Zhaoheng Ni, Sanjeel Parekh, Buye Xu, "ArrayDPS-Refine: Generative Refinement of Discriminative Multi-channel Speech Enhancement"
- **ICLR 2026**: Sattwik Basu, Chaitanya Amballa, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi, Romit Roy Choudhury, "Contrastive Diffusion Guidance for Spatial Inverse Problems"
- **WASPAA2025**: Yulun Wu, Zhongweiyang Xu, Jianchong Chen, Zhong-Qiu Wang, Romit Roy Choudhury, "Unsupervised Multi-channel Speech Dereverberation via Diffusion"
- **ICML2025**: Zhongweiyang Xu, Xulin Fan, Zhong-Qiu Wang, Xilin Jiang, Romit Roy Choudhury, "ArrayDPS: Unsupervised Blind Speech Separation with a Diffusion Prior"
- **NeurIPS2024 Workshop**: Zhongweiyang Xu, Debottam Dutta, Yu-Lin Wei, Romit Roy Choudhury, "Multi-Source Music Generation with Latent Diffusion"

- **ISCA INTERSPEECH2024:** *Zhongweiyang Xu, Ali Aroudi, Ke Tan, Ashutosh Pandey, Jung-Suk Lee, Buye Xu, Francesco Nesta*, “FoVNet: Configurable Field-of-View Speech Enhancement with Low Computation and Distortion for Smart Glasses”
- **IEEE ICASSP2024:** *Muqiao Yang, Chunlei Zhang, Yong Xu, Zhongweiyang Xu, Heming Wang, Bhiksha Raj, Dong Yu*, “uSee: Unified Speech Enhancement and Editing with Conditional Diffusion Models”
- **Information Fusion:** *Andong Li, Guochen Yu, Zhongweiyang Xu, Cunhang Fan, Xiaodong Li, Chengshi Zheng*, “TaBE: Decoupling spatial and spectral processing with Taylor’s unfolding method in the beamspace domain for multi-channel speech enhancement”
- **IEEE ICASSP2024:** *Zhongweiyang Xu, Yong Xu, Vinay Kothapally, Heming Wang, Muqiao Yang, Dong Yu*, “SpatialCodec: Neural Spatial Speech Coding”
- **SPEAR Workshop:** *Zhongweiyang Xu, Debottam Dutta, Xulin Fan, Mark Hasegawa-Johnson, Romit Roy Choudhury*, “Multichannel Speech Enhancement for SPEAR Challenge: A Three Stage Approach”
- **IEEE ICASSP2023:** *Zhongweiyang Xu*, Xulin Fan*, Mark Hasegawa-Johnson*, “Dual-Path Cross-Modal Attention for Better Audio-Visual Speech Extraction”
- **ACM ICML2022:** *Zhongweiyang Xu, Romit Roy Choudhury*, “Learning to Sparate Voices by Spatial Regions”

PREPRINTS

- **arXiv:** *Xilin Jiang*, Qiaolin Wang*, Junkai Wu*, Xiaomin He*, Zhongweiyang Xu*, et al.*, “AVMeme Exam: A Multimodal Multilingual Multicultural Benchmark for LLMs’ Contextual and Cultural Knowledge and Thinking” (ACL 2026 on submission)
- **arXiv:** *Debottam Dutta, Chaitanya Amballa, Zhongweiyang Xu, Yu-Lin Wei, Romit Roy Choudhury*, “Learning Energy-based Variational Latent Prior for VAEs”
- **arXiv:** *Heming Wang, Meng Yu, Hao Zhang, Chunlei Zhang, Zhongweiyang Xu, Muqiao Yang, Yixuan Zhang, Dong Yu*, “Unifying Robustness and Fidelity: A Comprehensive Study of Pretrained Generative Methods for Speech Enhancement in Adverse Conditions”

HONORS AND AWARDS

- **2018-2021:** UIUC ECE Highest Honor Graduation
- **2024-2025:** A. R. Buck Knight Fellowship

PROJECTS

- **Blind Image Deblurring with Flow Prior:** Use an image-domain flow-matching model as a prior, to solve blind image deblurring. We propose a flow posterior sampling technique with 2 different variants: 1) use a Plug-and-play method to estimate the blurring kernel during each sampling step, 2) train another flow model for the kernel, and then apply parallel diffusion sampling for both kernel and image.
- **Linux Kernel Project:** Develop an OS Kernel from scratch for ECE391 (about 10k lines of C and X86).
- **BeatDance on FPGA:** Develop a GuitarHero-like game on a Cyclone-V FPGA.
- **RISC-V CPU:** Use SystemVerilog to develop a RISC-V CPU with cache and pipeline.

SKILLS

- **Languages** Python, C, C++, SystemVerilog
- **Coursework and Skills** LLMs, MLLMs, Generative Models, Machine Learning, Deep Learning, Multimedia Signal Processing, Random Process, Generative AI, Computer Vision, NLP, Optimization, Adaptive Signal Processing, Information Theory, Speech Processing, Advanced DSP, Vector Space Signal Processing, Detection and Estimation Theory, Wireless Sensing, Digital System Design, Operating System.